**Q1.** Spots may be caused by bacteria in the skin. A researcher investigated the effect of spot-lotion on bacteria.

(a) He grew bacteria on the surface of jelly in a Petri dish.

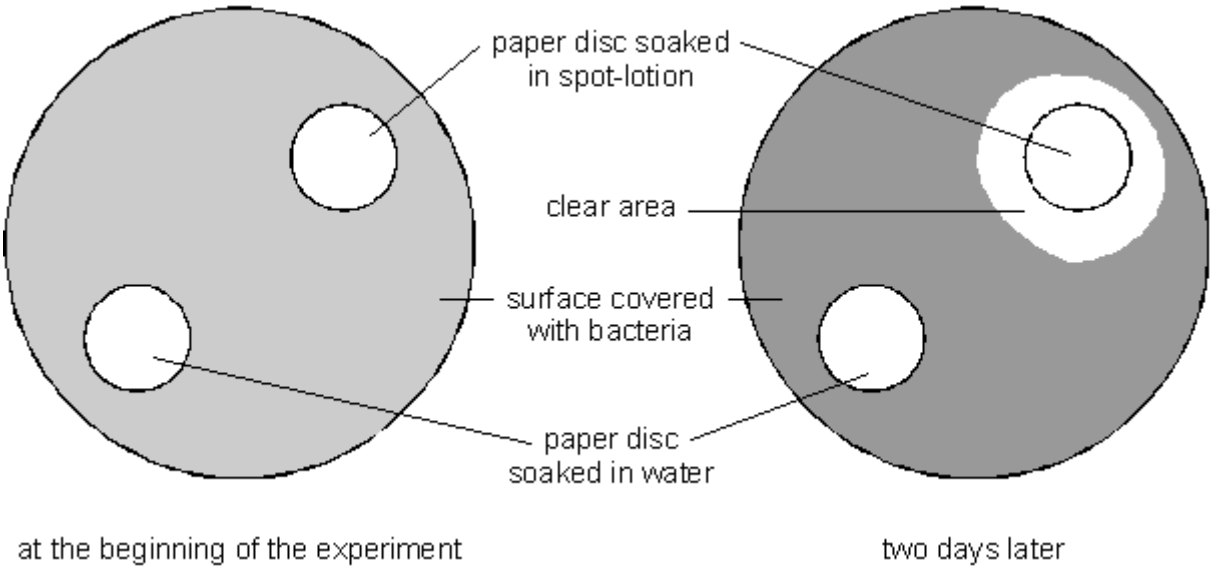
At what temperature would the bacteria reproduce quickly?

Tick the correct box.

| | | | |
|---|---|---|---|
| 100°C | ☐ | 4°C | ☐ |
| 37°C | ☐ | –15°C | ☐ |

1 mark

(b) The researcher placed two small paper discs onto the surface of the jelly.
One disc had been soaked in spot-lotion. The other disc had been soaked in water.
The diagrams below show the jelly at the beginning of the experiment and two days later.




Suggest what had happened to the bacteria in the clear area around the paper disc soaked in spot-lotion.

……

……

1 mark

(c) What was the control in this experiment?

……

……

1 mark

(d) Give **two** safety precautions the researcher should take to avoid contact with the bacteria.

1. ……

2. ……

2 marks
Maximum 5 marks

Mark schemes

## Q1.

(a) 37°C

*if more than one box is ticked, award no mark*

**1 (L5)**

(b) the lotion killed the bacteria

*accept 'they died' or 'they were killed **or** destroyed'*

**1 (L5)**

(c) the paper disc soaked in water

*accept 'the other disc'*

**1 (L5)**

(d) any **two** from

- keep the lid on the dish
- seal **or** secure the dish
- wear gloves
- wear a mask **or** goggles
- use tweezers to add the paper disc

*do **not** accept 'do the experiment in a fume cupboard'*

**2 (L5)**

**[5]**